

Luigi Gargioni

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WORK EXPERIENCE

PhD Student
Antares Vision [01/12/2022 – Current]

City: Travagliato, Brescia
Country: Italy

Telemedicine and Digitization of Health Services

Software Development Engineer - Freelance
Gargioni Luigi [12/2022 – Current]

City: Brescia
Country: Italy

• Full-Stack Software Development
• Website: https://luigigargioni.github.io/

Scoutman
TieBreakTech [29/09/2019 – Current]

City: Montichiari (Brescia)
Country: Italy

Scoutman for the Women's Serie A Volleyball League at PalaGeorge in Montichiari (BS) and PalaRadi in Cremona.

Assignments:
• Statistics collection and collaboration with press officers.

Software Engineer - R&D Developer
Kline s.r.l. [16/11/2019 – 30/11/2022]

City: Leno (Brescia)
Country: Italy

Kline develops and manufactures integrated software platforms, technological solutions and services for financial intermediaries.

Research & Development Department - Assignments:
- Reengineering existing application with functional programming paradigm and web technologies: JavaScript, TypeScript, Webpack and React (Redux, Styled Components, Jest, ...).
- Prodcut Owner of two outsourced projects, collaborating with an international company
- Recruitment

Programmer Analyst - Previously:
- Functional analysis of banking and financial procedures.
- Implementation of modules in Delphi, Java and PL/SQL (Oracle DB).
- Customer support activities in relation to the applications developed and distributed.

Collaborator
Punto Pizza [27/02/2011 – 29/12/2021]

City: Gambara (Brescia)
Country: Italy

Take-away and sliced pizzeria.

Assignments:
- Pizza chef help and counter attendant at the family business.

Software Engineer
C&B Servizi Informatici [29/11/2016 – 28/06/2017]

City: Casalmoro (Mantova)
Country: Italy

Internship outside the university carried out during the period of study: after the established period, possibility of recruitment not accepted for choice to continue the studies and start of the master's degree course in computer engineering. Occasional project work in 2020/2021.

Assignments:
- Software development and debugging in Microsoft an Web environment: Visual Basic .NET, Visual Basic 6, ASP:NET, Javascript, Microsoft SQL Server and Access.
- Installation and maintenance of local networks, departmental servers and network equipment.
- Supporting personnel in installation and maintenance of personal computers and peripherals both at our headquarters and at the customer's premises.

Bartender and waiter
Camping Village San Francesco [30/05/2014 – 29/08/2014]

City: Desenzano del Garda (Brescia)
Country: Italy

Assignments:
- Bartender and waiter for the summer season.

EDUCATION AND TRAINING

PhD Student - Technology for Health
University of Brescia [01/12/2022 – Current]

Address: Via Branze 38, 25123 Brescia (Italy)
Website: https://tech4health.unibs.it/

Qualification for IT engineer profession practice
University of Brescia [22/07/2020 – 22/07/2020]

Address: Via Branze 38, 25123 Brescia (Italy)
Website: https://www.unibs.it/

Master Degree, Computer Engineering
University of Brescia [2017 – 2019]

Address: Via Branze 38, 25123 Brescia (Italy)
Website: https://www.unibs.it/
Final grade: 107/110
Thesis: A domain-independent environment for user-oriented programming of collaborative robots
Link: https://www.sciencedirect.com/science/article/abs/pii/S073658452100106X

Bachelor Degree, Computer Engineering
University of Brescia [2012 – 2016]

Address: Via Branze 38, 25123 Brescia (Italy)
Website: https://www.unibs.it/
Thesis: Vulnerability checking of systems: general concepts and analysis of the OpenVAS tool

Technological Scientific High School
Blaise Pascal Higher Education Institute [08/2007 – 06/2012]

Address: Via Solferino 92, 25025 Manerbio (Brescia) (Italy)
Website: https://www.iis-pascal.edu.it/
Thesis: Concepts of nuclear chemistry and analysis of Chernobyl and Fukushima disasters

LANGUAGE SKILLS

Mother tongue(s): **Italiano**

Other language(s):

Inglese	**Francese**
LISTENING B2 **READING** B2 **WRITING** B2	**LISTENING** A1 **READING** A1 **WRITING** A1
SPOKEN PRODUCTION B2 **SPOKEN INTERACTION** B2	**SPOKEN PRODUCTION** A1 **SPOKEN INTERACTION** A1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

DIGITAL SKILLS

Front-End React Redux JavaScript Bootstrap HTML5 CSS3 / Node.Js / HMI (Human Machine Interface) / R&D / Robotics / SVN, Git, GitLab

PUBLICATIONS

A hybrid approach to user-oriented programming of collaborative robots
[2022]

Robotics and Computer-Integrated Manufacturing, Volume 73, February 2022

The research reported in this paper proposes a new approach to collaborative robots that aims at improving the simplicity and efficiency of the programming task for non-technical users. It is grounded on three standpoints: (i) an elementary and disciplined paradigm for robot programming, called the simple programming journey, (ii) a hybrid interaction mode where robot tasks can be programmed using a natural language chat and, if necessary, can be completed and finalized through a block-based interface, and (iii) a robust cognitive match between the mental models of the user and the programming interface. The proposed approach has been implemented and tested through the development of a prototype programming environment called CAPIRCI, which can be tailored to different application domains through the definition of objects, locations, and actions. CAPIRCI has been tested by real users with a COBOTTA robot by DENSO WAVE Ltd. Two experimental tests have been carried out in order to validate the novel approach proposed and to assess its impact on end-user programming. The results obtained show that a hybrid approach exploiting both natural language dialogue and block-based interaction can help make the programming task easy and efficient for non-technical users.

Human-robot collaboration in healthcare: new programming and interaction techniques
[2023]

IS-EUD 2023, 9th International Symposium on End-User Development, 06-08 June 2023, Cagliari, Italy

In the field of healthcare, collaborative robots can be utilized to enhance productivity and efficiency. This research will delve into the development and investigation of new techniques to enable effective interaction with collaborative robots. The focus will be on designing innovative techniques that can be easily understood and implemented by individuals with no technical knowledge in computer science or robotics. Collaborative robots can be used to automate repetitive tasks, allowing healthcare professionals to focus on more complex procedures. For instance, by leveraging this technology, it is possible to significantly increase the efficiency and

speed of therapy preparation, ultimately improving patient outcomes. The study will explore various approaches to simplify and streamline the programming process, reducing the need for technical knowledge and expertise.

Can ChatGPT Support End-User Development of Robot Programs?

[2023]

CoPDA 2023, Cultures of Partecipation in the Digital Age, International workshop @ IS-EUD 2023

In this paper, we investigate the use of OpenAI ChatGPT to improve the natural language understanding of an End-User Development environment, called CAPIRCI, supporting users neither expert in computer programming nor expert in robotics to create programs for a collaborative robot. The integration of ChatGPT in CAPIRCI is studied to be transparent for the user, who will be allowed to check and correct that program description generated by ChatGPT, by interacting with an intuitive block-based interface, according to a Human-Centered Artificial Intelligence design approach.

Designing Human-Robot Collaboration for the Preparation of Personalized Medicines

[2023]

GoodIT 2023, International Conference on Information Technology for Social Good, Lisbon, Portugal

Advancements in robotics and automation technologies have the potential to enable breakthrough innovations in a variety of industries, and the pharmaceutical sector is no exception. The preparation of galenic formulations, involving the compounding and dispensing of medications, when personalized medicines are needed, e.g., to overcome allergy problems, is a critical process in the field of small scale pharmaceutical manufacturing. Traditionally, this process has relied solely on human expertise of pharmacists and their manual labor, which can be time-consuming, prone to errors, and subject to variations in quality. To overcome these limitations, the use of collaborative robots is envisaged in our project. A collaborative robot can in fact work with the pharmacist synergistically, by improving accuracy and increasing productivity. However, the main challenge is providing the pharmacists with an interactive system that supports them in robot programming. In this paper, we analyze the problem from the users' point of view and propose preliminary low-fidelity prototypes of an interactive system suitable to pharmacists' needs and skills.

A Systematic Review on Pill and Medication Dispensers from a Human-Centered Perspective

[2023]

Journal of Healthcare Informatics Research, Springer - Peer Review Stage

As medication adherence represents a critical challenge in healthcare, pill and medication dispensers have gained increasing attention as potential solutions to promote adherence and improve patient outcomes. Following the PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) methodology, we carried out a systematic literature review on papers indexed in Scopus, which present solutions for pill or medication dispensers. Given the importance of user acceptance for these solutions, the research questions of the survey are driven by a human-centered perspective. We first provide an overview of the different solutions, classifying them according to their stage of development. We then analyze each solution considering its hardware/software architecture. Finally, we review the characteristics of user interfaces designed for interacting with pill and medication dispensers and analyze the involvement of different types of users in dispenser management. On the basis of this analysis, we draw findings and indications for future research that are aimed to provide insights to healthcare professionals, researchers, and designers who are interested in developing and using pill and medication dispensers.

PROJECTS

Gestionale per carpenteria metallica

[12/2020 – 08/2022]

Management software for metalworkers developed with web technologies:

- Front-end: Typescript, Electron, React, Redux, Webpack, Ant Design, Styled-Components
- Back-end: Typescript, Node.js, Express, Sequelize, Morgan, JWT, bcrypt
- Database: MariaDB
- Versione control with Git (GitHub)
- Support (HW/SW): Raspberry Pi, GitHub Releases, Husky, PM2

CAPIRCI (Chat And Program Industrial Robots through Convenient Interaction)

[02/2019 – 10/2019]

A multi-modal web application supporting end users, with no expertise in computer science, to define and modify tasks to be executed by collaborative robots. The application provides two interaction modalities, the former based on a chat interface, the latter presenting a visual programming language inspired to block-based solutions but tailored to the domain at hand.

Technologies used: Python, Django, OpenCV, Javascript, Ajax, Css, Standford Parser, ORiN and B-cap.

Robot used: Denso Cobotta with Canon N10-W02 camera and Denso VS.

Link: https://www.sciencedirect.com/science/article/abs/pii/S073658452100106X

PallavoloRemedello.it

[08/2018 – 05/2019]

Web site for the volleyball team A.S.D. Pallavolo Remedello.

Technologies used: Javascript, jQuery, Ajax, PHP and CSS.

HONOURS AND AWARDS

MongoDB Node.js Developer Path

MongoDB [02/2023]
Link: https://learn.mongodb.com/learn/learning-path/mongodb-nodejs-developer-path

Postman API Fundamentals Student Expert

Postman [01/2023]

Credential ID: WIoUhlzfQOqVq57BMYKK_Q

Link: https://api.badgr.io/public/assertions/WIoUhlzfQOqVq57BMYKK_Q?identity__email=l.gargioni%40studenti.unibs.it

Machine Learning

MIT Open Learning [08/2022]
Link: https://openlearninglibrary.mit.edu/courses/course-v1:MITx+6.036+1T2019/course/

Artificial intelligence and sustainability

Fastweb Digital Academy [04/2022]

Credential ID: 63382067693d40f6c9f071c284da76a972873f62

Link: https://learn.fastwebdigital.academy/badges/badge.php?hash=63382067693d40f6c9f071c284da76a972873f62

How to Write and Publish a Scientific Paper (Project-Centered Course)

École Polytechnique [04/2022]

Credential ID: 46CKHJA5BZRE

Link: https://www.coursera.org/account/accomplishments/verify/46CKHJA5BZRE

Elements of AI

University of Helsinki [11/2021]

Credential ID: 7kas3px5o4q

Link: https://certificates.mooc.fi/validate/7kas3px5o4q

Machine Learning
Stanford University [09/2020]

Credential ID: G5J6SZ2UXRDX

Link: https://www.coursera.org/account/accomplishments/verify/G5J6SZ2UXRDX

Basic Certification
3CX [09/2020]

Credential ID: ezftPKvV8p

Link: https://customer.3cx.com/documents/CertificationSharing.aspx?c=ezftPKvV8p

Cisco Networking Academy
Cisco Systems [04/2020]

- Intro to IoE
- Introduction to Cybersecurity
- Cybersecurity Essentials
- NDG Linux Essentials
- NDG Linux Unhatched

Link: https://www.youracclaim.com/badges/a4cdaf40-7ae7-4422-a89b-5b9ca09f7836

ISE I Integrated Skills in English - B1 of the CEFR
Trinity College London [03/2012]

European Computer Driving License (ECDL)
AICA - Italian Association for Computing and Automated Computing [06/2011]

HOBBIES AND INTERESTS

Volleyball

Football

Softair

Reading

NETWORKS AND MEMBERSHIPS

Associazione Alumni Università degli Studi di Brescia
[Brescia, 11/2022 – Current]
Link: https://alumni.unibs.it/documents/badges/badge.php?hash=02608da02a1dbd8ef89a8bd5b40403df2a766f75